

AI Grinding for Fun and Cryptanalysis

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Abstract

We present an autonomous cryptanalysis workflow in which agents generate, test, and refine hypotheses before human review. The autonomous stage returns reproducible candidate findings with exact witnesses, controls, code, environments, and run records. A researcher then determines whether the evidence establishes a break, defect, or coverage gap. The attacks below serve as case studies of this method.

Two failure modes recur, and naming them is more useful than listing the targets, so they come first. In the first, a public algebraic map or an input representation erases or exposes the relation a construction must keep hidden. The maps are unglamorous: multiplication by zero, the boundary coefficients of a polynomial product, a quotient, a character, a Schur square, or a variable-length byte encoding without boundaries. In the second, a distribution is not the law it is claimed to be, whether a simulator, an error law, or a parameter certification. Several targets fail on both axes at once, which is what makes them decisive.

Every result carries an exact witness and at least one discriminating control, and every stated boundary carries a proof. Three further targets yielded no attack but a narrower guarantee than a generic reading suggests, and are reported separately so that no coverage gap reads as a break.

Eight published constructions fail at parameters or claims their own authors state. A Ring-LWR commitment opens to every message with probability one; two middle-product encryption rows are decrypted from a single ciphertext; a lattice e-voting protocol loses receipt-freeness, so a coerced voter can no longer hide how they voted; a published permutation recovery against an updatable encryption scheme lifts by linear algebra to the old decryption key; an explicit normal basis collapses a degree-63 instance into seven degree-nine ones; a signature hash outside

the lattice setting maps two printable messages of equal length to one digest; and a rerandomisable scheme’s accept bit is a threshold oracle on its own decryption noise. Kept separate on purpose, a group-ring decision claim and a multivariate MinRank hardening fail as assumption and accounting defects rather than as broken constructions. Every failure sits one level above the assumption it rests on. Who knows what else awaits out there?

Keywords: cryptanalysis, lattice-based cryptography, multivariate cryptography, machine-assisted research, reproducible evidence

1 Grinding: a direction for machine-assisted cryptanalysis

AI may offer many ways of taking a construction apart. A model reads a paper and returns candidate lines of attack in quantity, and almost none of them survive contact with an exact test. Deciding which candidate is real is the expensive part, and that is the subject here.

Grinding is the name for the arrangement that answers it. Candidate generation is treated as deliberately high in volume and low in precision, which is affordable only because the workflow never treats generator output as evidence. Exact arithmetic, a discriminating control, and the source document decide what is admitted. Whatever survives is pushed until it either breaks something the paper promises or runs into a wall that can be proved to be a wall. Cryptanalysis has always rewarded depth, one person carrying one idea a long way, and nothing here replaces that.

The snail is an emblem, not a claim about pace. What it stands for is the trail: ground covered one claim at a time, in the open, where anyone can retrace it. Whether this amounts to a new direction in cryptanalysis and in mathematics more broadly is an open question. What follows is the yield and the method that produced it.

The blunt version: pick a paper, read what it actually claims, attack every step where the argument moves from one setting to another, turn each suspicion into a precise statement, test that statement on exact small instances, discard bad ideas quickly, and push whatever survives until it either stops for a reason that can be written down or breaks the advertised security property. Then check everything against the source and record what broke and what did not.

The push is where the results come from. Auditing a proof once is ordinary. Every surviving result carries a stated limitation, and that limitation is the next thing to attack:

$$\begin{aligned} \text{idea} &\rightarrow \text{exact witness} \rightarrow \text{“only works when ...”} \\ &\rightarrow \text{attack that limitation} \rightarrow \text{stronger result.} \end{aligned}$$

That move has two productive exits, so it is never wasted. A boundary that moves gives a stronger attack. A boundary that refuses to move becomes a maximality statement, which is also a result. Six of the findings below reach the top level and break an advertised security property; the rest stop lower, and Table 1 records where

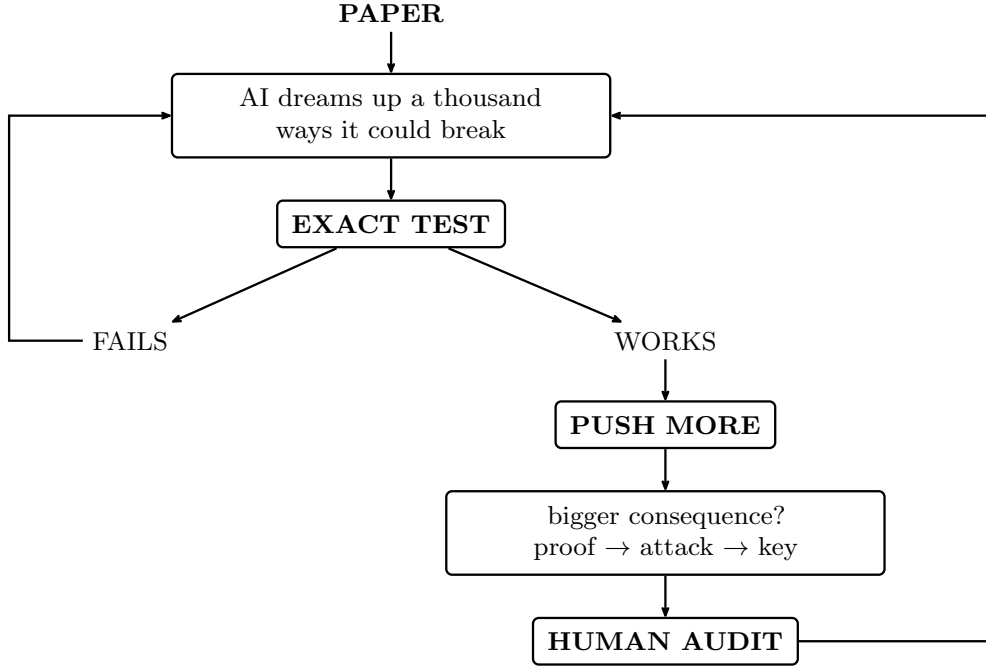


Fig. 1 Most ideas fail. Survivors are pushed until they either hit a proved boundary or break the advertised security property. Everything that survives is rechecked against the exact paper version and an exact calculation or executable witness. The autonomous path produces a reproducible candidate result; human review determines its cryptanalytic status.

each one stopped and why. Nobody camped on level one for lack of trying. This is the operation that separates the method from ordinary machine-assisted auditing. Auditing asks whether a candidate flaw is real and stops when it has an answer. Grinding treats every verified limitation of a real flaw as the next candidate, so a confirmed result is an input rather than an endpoint. The arrangement works as a game with levels, which is what Figure 2 shows: a finding enters at the left and climbs for as long as the evidence will carry it.

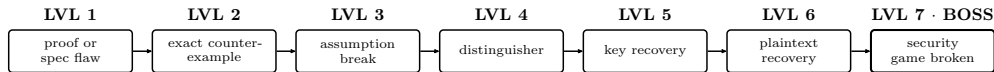


Fig. 2 The game. Every finding enters at level one and climbs for as long as the evidence carries it, and it stops only when the next level cannot be reached: either the bridge to a real consequence fails, or the wall ahead is proved to be a wall. Table 1 records the level each finding here reached.

Generation needs direction, and the direction has two parts. The first is where to look. A paper is attacked at its joins: a change of algebra, a representation or encoding change, a distribution carried through a map, a rounding or CRT conversion, a quotient, an exceptional value such as zero or a nonunit, a quantifier that widens from one object to many, and a parameter substituted into a script instead of into the

Target	Join attacked	Axis	First survivor	Limitation pushed	Level
Ring-LWR	Exceptional value inside an accepted set	A	$d = 0$ opens the zero commitment to every message	Binding only, so the three protocols were attacked next	7
Spinel	Representation change from bytes to walk symbols	A	Variable-length codewords concatenated without boundaries	One pair only, so count the bytes and lengths it reaches	7
Middle-product	Ordinary product used where a wrapping one is assumed	A, B	Boundary coefficients peel the masks at $t = 9$	Published bound vacuous from $t = 16$, so branch instead of abort	7
CRT-RLWE reaction	Exact noise extraction meeting a validity signal	A	Coherent shifts cancel until the centre lift wraps	One coefficient, so binary-search the whole polynomial	7
MinRank hardening	Rank raised by perturbation, then averaged	A	The aa block is publicly computable	Average is not the attacker's direction, so take the tail	1
E-voting	Distribution carried through a simulator	B	Simulator marginal differs from the genuine law	Distance looked parameter-dependent, so bound it over all p	7
Hollow-LWE	Quotient, on top of a published recovery	A, B	$W_{ij}V_{ij}^{-1} = d_id_{I_j}$	Signs fixed only per component, so use the public key relation	7
Function-field	CRT decomposition of an explicit basis	A, B	Galois orbit supported factorwise	Projected rate might rise, so compute it	5
Group ring, full	Quantifier widened from one ring to a family	A	Augmentation is a public map onto $\mathbb{Z}$	One functional only, so count every sign character	3
Group ring, quotients	Quotient of the selected families	A, B	Rank-four maps onto $M_2(\mathbb{F}_q)$	Only the quotient image, so try to widen it	3
Hollow-LWE parameters	Parameter put into a script, not into its theorem	B	Estimator invoked at k , not $k - h$	Says nothing about the true level, so cost the repair	1

Table 1 The trace each result actually followed, from the join that was attacked to the level of Figure 2 at which it stopped. The limitation column is the push: in every row it is a statement the first survivor could not support, taken as the next hypothesis. Two of these pushes ended in a proved boundary instead of a stronger attack, which Lemma A.11 counts as the second productive exit.

theorem it came from. Every finding below sits on one of those, and assembling that list is the cheap part of the work.

The second part is what to ask on arrival, and two questions carry almost all of it. Which efficiently computable map survives this join with useful signal still in it, and which distribution actually results from it. Those two questions are the axes of Section 3, which is why that section appears before the findings and not as a summary of them. Both are cheap to ask, neither requires understanding a construction in full, and both can be put to a generator that has read one paper and nothing else.

Table 1 records the trace each result followed, so that the findings can be read as instances of the method rather than as work labelled with it afterwards. Section 4.3 is the clearest case of pushing. Stopping at “the attack works at $t = 9$ and the theorem starts at $t = 98$ ” would have been correct and uninformative. Attacking that boundary showed first that the published failure bound measures the wrong quantity and the attack in fact runs to $t = 24$, and then that 98 is essentially the whole-ciphertext entropy threshold 96.32 plus the security slack $2\lambda/(k + 1)$. The requirement is not an artefact of one reduction; it is the point at which the ciphertext stops determining the masks.

Autonomous search and human review.

A large language model reads the sources, enumerates candidate paths from a claim to a consequence, proposes witnesses, and writes the checking code. Such proposals are unreliable in a specific and predictable way: they are fluent, frequently wrong in the direction of overclaiming, and they do not separate a proof gap from a break unless forced to. The working rule is that no proposal counts as evidence until it is reduced to an exact statement a short deterministic program can check, alongside a control whose outcome was fixed before it ran. Several plausible candidates fail there, and one appears in Section 4.9 as a rejected strengthening because its failure is informative. The division holds throughout: the machine supplies breadth of conjecture, the exact machinery decides admissibility, and consequence, scope, and provenance stay with the researcher.

Autonomous search proposes and tests candidate findings; human review determines their cryptanalytic status.

Admission depends on exact checks and predetermined controls rather than on generator confidence. Changing the generator therefore affects search yield, but not the admission criterion. Machine assistance makes it possible to pose the two questions across many papers and test candidate answers autonomously. Appendix A states the same arrangement as a calculus.

The autonomous stage returns the claim under test, source version, witness, control, code, execution environment, run record, known limitations, and proposed classification. That handoff is a reproducible candidate result, not a classified one.

Three disciplines do the filtering. Every equality, modular, integer, encoding, and bound predicate is evaluated in exact rational or integer arithmetic; the binomial tails in Section 4.4 are exact rationals over 27^{4096} , not normal approximations. Every executable check carries a control whose expected outcome is fixed in advance, listed in Table 7. Breaks, assumption defects, and coverage gaps are recorded as different things and never merged.

One failure shows what the discipline exists to catch. The first end-to-end run of the group-ring distinguisher did not separate genuine from uniform samples: at a convenient small modulus the union bound evaluates to 1.78, so no separation is available there. The analysis was right and the run did not support it. Rerunning at the parameter sequence the analysis actually specifies gives complete separation. Any reported reproduction not run at the stated parameters should be treated as unverified.

<i>Assumptions and problems</i>			
LWE	Learning With Errors	RLWE	Ring Learning With Errors
MLWE	Module Learning With Errors	LWR	Learning With Rounding
MP-LWE	Middle-Product LWE	MP-CLWR	Middle-Product Computational LWR
LPN	Learning Parity with Noise	SIS	Short Integer Solution
HFE	Hidden Field Equations	MinRank	the problem of finding a low-rank combination of public matrices
<i>Constructions and notions</i>			
PKE	Public-Key Encryption	UPKE	Updatable Public-Key Encryption
ABE	Attribute-Based Encryption	AEAD	Authenticated Encryption with Associated Data
IND-CPA	Indistinguishability under Chosen-Plaintext Attack	IND-CR-CPA	the corresponding notion for ciphertexts predating a key update
<i>Other</i>			
CRT	Chinese Remainder Theorem	SLH-DSA	Stateless Hash-Based Digital Signature Algorithm
FORS	Forest Of Random Subsets, a component of SLH-DSA	WOTS+	Winternitz One-Time Signature Plus

Table 2 Abbreviations, expanded here and at first use in the text.

Costs quoted below are arithmetic operation counts, not wall-clock benchmarks, unless stated otherwise.

A procedure of this kind invites one obvious objection, and it is the reason Section 5 exists. Output consisting entirely of breaks would be indistinguishable from a selection procedure with an unknown filter on it, particularly when the generator is fluent and inclined to overclaim. The answer is to record the whole distribution of outcomes and keep the classes apart: attacks, assumption defects, coverage gaps, results belonging to other authors, paths left unresolved, and strengthenings that were tried and failed. Three targets produced no attack and are reported as such. One strengthening was refuted and is kept for the boundary its refutation proved.

Which leaves the claim the arrangement rests on. Hypothesis generation and cryptanalytic evidence do not need the same reliability. Generation can be made deliberately high in volume and low in precision, provided an exact and adversarially controlled admission boundary sits between a proposal and evidence, and provided the limitations of every survivor are attacked in turn. Generator quality determines search yield; exact checks and controls determine what survives. The machine does not have to be reliable for the research to be.

2 Results at a glance

Table 2 expands the abbreviations used throughout, and Table 3 lists every result with its status and the number that carries it. The derivations follow in the same order.

Target	Claim defeated	Status	Evidence
Ring-LWR commitment and proofs [16]	Binding, and soundness of all three protocols	Break, probability one	Zero multiplier opens one commitment to every message; extraction is undefined for half the challenge pairs
Spinel signature hash [15]	Collision and second-preimage resistance of the hash	Break, deterministic, no search	Distinct messages share one walk string; 244 of 255 nonzero bytes have an immediate second preimage
Two-limb CRT-RLWE rerandomisation [9]	Composed malicious-rerandomiser claim	Break, conditional on an observable reaction	Coherent CRT shifts make the accept bit a noise threshold; 131072 reactions give the key
HFE with LL' perturbations [8]	The MinRank hardening behind the parameter tables	Accounting and transcript defects	Ambient dimension publicly reducible, averaged rank attacker-chosen, and a degree-2 transcript invariant
Middle-product PKE, $t = 9$ rows [3, 14]	IND-CPA confidentiality	Break, passive, one ciphertext	All 2313 masks and the plaintext recovered, success ≥ 0.999999730074 ; range extends to $t = 24$
Lattice e-voting receipts [10]	Receipt-freeness and vote privacy against a transcript coercer	Break, unconditional	Simulator distance $\geq 1/15$ for every permitted parameter; test advantage 0.999999804160, and provably optimal
Hollow-LWE UPKE [1]	IND-CR-CPA confidentiality of pre-update ciphertexts	Break, given the published permutation recovery	A valid old key recovered and the challenge decrypted 12/12 at three dimensions, in $O(k^3 + nk^2)$ operations
Function-field Normal Ring-LPN [7]	Search hardness of the explicit constructive basis	Break of the printed instance	$X^{63} + X^7 + 1$ splits into seven degree-nine factors; 3584 candidate tests replace 2^{63}
Semidirect group-ring LWE [13]	Decision hardness, full rings and selected quotients	Assumption defect	Two samples distinguish with false acceptance $64N^{-7} \log^2 N$; quotient image shown maximal
Hollow-LWE parameters [1]	The printed 128, 192 and 256-bit labels	Certification defect	The estimator is called at dimension k , not $k - h$; repairing both defects costs 1.24 to 1.54×
Hint-MLWE [12]	The unrestricted parameterised definition	Assumption defect	A permitted degenerate choice gives advantage $1 - q^{-nm}$

Table 3 Every result, its status, and the evidence carrying it. Section 5 adds three targets that produced no attack.

How results are classified.

Three categories are kept apart throughout, because merging them would misstate the evidence. A *break* reaches a security property the source claims: key or plaintext recovery, binding, soundness, or a named privacy notion. An *assumption or certification defect* shows that a stated claim is false or unsupported without exhibiting an attack on a construction. A *coverage gap*, collected in Section 5, is a reading of what a source actually proves, not a defect in it. Results attributable to other authors are marked at the point of use and are never counted here.

3 Two axes: what to ask at each join

The two axes below are search directions before they are a classification. They are the questions put at every join, and they are what a generator is pointed at when it reads a construction for the first time. The constructions analysed here come from different lines of work and rest on different assumptions, and the same two questions apply to all of them without modification, which is the property that makes the search worth running at volume.

The axes also classify every defect found, and that is worth stating carefully. A search that only asks two questions will only return answers to those two questions, so the classification is a property of the search and not a discovery about lattice cryptography. What the classification does establish is narrower and still useful: neither question was exhausted by a single target, and both kept producing across unrelated constructions.

Axis A: a public map that should not have survived.

A construction hides a relation and publishes objects derived from it. If some efficiently computable map sends the published objects somewhere the hidden relation is either trivial or enumerable, the assumption is not doing the work it appears to do. The maps here are unglamorous: multiplication by zero, reading the first and last coefficients of a polynomial product, reduction modulo a group-theoretic ideal, summing group-ring coefficients, and squaring a code coordinatewise. The same axis includes representation changes: a non-uniquely-decodable input encoding identifies distinct messages before the advertised algebraic problem is reached. None requires lattice reduction.

Axis B: a distribution that is not the claimed law.

A construction, a simulator, or a parameter script asserts that some object follows a distribution. If the asserted law and the actual law differ, the consequences range from a distinguisher to an invalid security label. The mismatches here are a simulator whose marginal is wrong at every permitted parameter, an error law that a decision definition requires but never supplies, a projected noise rate that fails to rise when the dimension drops, and a lattice estimator invoked at the wrong source dimension.

Table 4 places each target on both axes. Several sit on both, which is why they are decisive: a public map exposes an object, and the distribution of that object is not what the proof needs.

4 Findings

Findings are ordered by demonstrated impact: affected scope, consequence, centrality to the assurance claim, and evidence strength.

4.1 A Ring-LWR commitment opens to every message

The commitment verifier accepts an opener-chosen multiplier whose permitted values include zero, so one commitment opens to two distinct messages for every public key.

Target	Axis A: the public map	Axis B: the distribution
Ring-LWR commitment	Multiplication by the permitted $d = 0$	Not load-bearing here
Spinel signature hash	Byte-to-ternary encoding, non-injective under concatenation	Not load-bearing here
Two-limb CRT-RLWE	Coherent shift surviving the CRT recombination	Centre-lift wrap moves the plaintext by exactly one
HFE with LL' perturbations	Principal aa block, and the kernel of a transcript relation	Differential rank taken at its average, not its tail
Middle-product PKE, $t = 9$	Boundary coefficients of an ordinary polynomial product	Binary masks make each layer a 2^t subset sum
Lattice e-voting receipts	Public special decryption of the claimed vote	Simulator marginal differs from the genuine convolution law
Hollow-LWE UPKE	Schur square, then a signed-graph traversal	Estimator invoked at dimension k instead of $k - h$
Function-field Normal Ring-LPN	Projection onto one CRT component	Projected Bernoulli rate unchanged, not amplified
Semidirect group-ring LWE	Augmentation, and rank-four quotients	Decision definition supplies a set, not a distribution

Table 4 Each target placed on both axes.

The construction of Talapatra, Mishra, and Mukhopadhyay [16] commits to m by hashing and rounding,

$$\mathbf{c} = \text{Round}_{q \rightarrow p_1}(\mathbf{a}H(\text{enc}(m) \parallel \text{enc}(r))).$$

The honest opening uses multiplier 1, but verification accepts any (m, r, d) with

$$d \text{ Round}_{q \rightarrow p_1}(\mathbf{a}H(\text{enc}(m) \parallel \text{enc}(r))) = \mathbf{c}, \quad \|d\|_\infty \leq 1, \quad \deg d < n/2.$$

The accepted set contains $d = 0$. Taking $\mathbf{c} = 0$ and $d_0 = d_1 = 0$ makes both verification equations read $0 = 0$ for any two distinct messages, independently of the hash outputs and the public key. Binding fails with probability one at every valid parameter set.

The same cancellation defeats the proof system. In the preimage protocol, a prover sends $\mathbf{c}_{\text{aux}} = 0$, receives the challenge δ , and answers with $\mathbf{t} = -\delta\mathbf{c}$ and $s_m = 0$, opening the auxiliary commitment with multiplier zero. Both verifier equations become $0 = 0$ for every challenge, so the failure is in soundness, not in extraction alone. The two derived protocols are undefined as printed, since they add elements of R_p^k to elements of $R_{p_1}^k$; after the natural type correction they accept explicit false statements under the same strategy. Independently, the special-soundness extractor divides by challenge differences such as $x - 1$, which is a zero divisor in $\mathbb{F}_2[x]/(x^{512} + 1)$ because $x^{512} + 1 = (x + 1)^{512}$ there. That example is not isolated. Reduction modulo two lands

in the local ring $\mathbb{F}_2[x]/((x+1)^{512})$, so an element is a unit exactly when its modulo-two weight is odd; challenges are binary, so the difference is a unit exactly when the Hamming distance is odd. The fraction of distinct challenge pairs at even distance is $(2^{255} - 1)/(2^{256} - 1) = \frac{1}{2} - 1/(2^{257} - 2)$, so an extractor drawing two independent uniform challenges divides by a nonunit with probability one half. That is a constant, not a negligible quantity, and repetition does not repair it.

The predecessor Ring-LWE commitment [6] inverts its opening factor and therefore excludes zero. Ring-LWR hiding is untouched by this attack; what fails is binding, and the soundness of everything built on it.

4.2 A signature hash collides before its hard problem is reached

An encoding that converts bytes to walk symbols by suppressing leading zeros is not injective, so two distinct messages drive the same algebraic walk and produce the same digest with no search.

Spinel [15] keeps the SLH-DSA architecture and replaces its hash with an algebraic one, whose security rests on navigating a Cayley graph of $\text{SL}_4(\mathbb{F}_p)$. That problem is never reached. Each byte $b \in \{0, \dots, 255\}$ is written in base 3, which needs six trits since $3^6 = 729 > 256$; the specification then suppresses the leading zeros, maps each remaining trit t to $t + 1$, and concatenates the resulting codewords $C(b)$ into the string $E(M) = C(b_1) \parallel \dots \parallel C(b_m)$ that drives the walk.

Every byte does have a unique minimal representation, and that is the step the argument turns on: the source transfers the underlying hash's security on the ground that it applies only one-to-one encodings to the input and output. Per byte the code is indeed injective. A concatenation of variable-length codewords without boundaries is not. Since $0\text{x}01 = (1)_3$ and $0\text{x}04 = (11)_3$,

$$E(01\ 04) = 2 \parallel 22 = 222 = 22 \parallel 2 = E(04\ 01),$$

so the walk receives one string for two messages and every later matrix operation agrees. The witness is printable and length-preserving as well: $0\text{x}28 = (1111)_3$ and $0\text{x}79 = (11111)_3$ give $E("\text{y}") = E("\text{y}(") = 2^9$. No group element is computed to find either pair, and neither depends on p , on the generators, or on the output serialisation.

Second preimages follow with the target fixed first. The codeword for $0\text{x}04$ is 22, and $01\ 01$ encodes to $2 \parallel 2$, so $01\ 01$ is a second preimage of the one-byte message 04 . A nonzero byte admits this splitting unless its minimal representation has a single nonzero trit, which happens only for the eleven values $d \cdot 3^j$ with $d \in \{1, 2\}$, so 244 of the 255 nonzero values have an immediate second preimage. Length is not a defence either. Exhaustive enumeration of the $255^2 = 65025$ two-byte strings over nonzero bytes gives 49963 distinct encodings, with 26730 inputs in nontrivial collision classes; reading a uniform 64-byte value as 32 two-byte blocks, at least one block admits a same-length replacement with probability at least $1 - (1 - 26730/65536)^{32} = 0.999999948$. Multicollisions are immediate, since $"(\text{y}"$ and $"\text{y}("$ may be substituted independently at r positions to give 2^r equal-length messages with one digest.

The source instantiates every call to the tweakable hash $T_h(P, T, M) = H(P\|T\|M)$ by passing the byte string $B = P\|T\|M$ to its hash, so $E(M_0) = E(M_1)$ gives

$$H(P\|T\|M_0) = H(P\|T\|M_1)$$

for every fixed prefix and tweak.

Whether that reaches a signature forgery turns on one function the source does not pin down. In SLH-DSA the message digest H_{msg} is not a tweakable-hash call, and the specification here fixes only the tweakable hash. If H_{msg} is instantiated with the same primitive, a chosen-message adversary requests a signature on M_0 and returns it unchanged with M_1 : verification derives the same digest and walks the same FORS, WOTS+, and hypertree path, so the substitution succeeds with probability one. If H_{msg} retains a conventional hash, the two messages give different digests and different indices, and this substitution fails. The collision and second-preimage results above hold either way.

The repair is the construction the paper states before it optimises. Retaining all six trits fixes the boundaries and restores injectivity, and any uniquely decodable byte code does the same. Prefixing the total message length does not, because the witnesses above already have equal lengths.

A second and unrelated defect sits in the parameter argument. The digest is sixteen 32-bit words holding the entries of a matrix of $\text{SL}_4(\mathbb{F}_p)$ with $p = 2^{31} - 1$, described as 2^{512} possible values. The top bit of each word is always zero and the determinant is constrained, so with $|\text{SL}_4(\mathbb{F}_p)| = p^6(p^2 - 1)(p^3 - 1)(p^4 - 1)$ the digest space is $\log_2 |\text{SL}_4(\mathbb{F}_p)| = 465.000$ bits, and the idealised classical birthday scale is $2^{232.5}$. Repairing the encoding does not change this.

Not claimed. First-preimage resistance: what fails is the injectivity of E , and given an arbitrary target matrix an attacker must still find a walk reaching it. Nor is the $\text{SL}_4(\mathbb{F}_p)$ navigation problem attacked, classically or quantumly. The message substitution above is claimed only under the stated condition on H_{msg} , and no implementation was available to test it.

4.3 Middle-product encryption is decrypted from one ciphertext

At both published $t = 9$ parameter sets, the boundary coefficients of the first ciphertext component recover all 2313 binary encryption masks, and the plaintext follows by public computation.

The MP-LWE scheme of Roşca, Sakzad, Stehlé, and Steinfeld [14] and the MP-CLWR scheme of Bai et al. [3] both publish

$$c_1 = \sum_{i=1}^t r_i a_i \pmod{q},$$

where each r_i has binary coefficients. This is ordinary, non-cyclic polynomial multiplication, which is triangular at both ends. The constant coefficient of c_1 depends only on

the constant coefficients of the r_i ; once those are removed, the next coefficient exposes the next layer. After subtracting the earlier layers, layer ℓ satisfies

$$T_\ell = \sum_{i=1}^t a_{i,0} r_{i,\ell} \pmod{q},$$

so a single public table of 2^t subset sums decodes every layer. The highest-degree coefficients give an independent reverse recurrence built from $(a_{i,n-1})_i$.

Injectivity fails only if some nonzero $d \in \{-1, 0, 1\}^t$ has $\sum_i a_{i,0} d_i = 0$. The vectors d and $-d$ define the same event, so there are $(3^t - 1)/2$ sign classes, and in each the chosen nonzero coordinate is ± 1 , hence a unit modulo every q . The two endpoint tables depend on disjoint independent coefficient vectors, so both fail with probability at most ε^2 where $\varepsilon = (3^t - 1)/(2q)$. At the published $t = 9$ and $q = 18941623$,

$$\varepsilon = \frac{9841}{18941623}, \quad 1 - \varepsilon^2 = 0.999999730074 \dots,$$

with tables of only 512 entries each.

For MP-LWE the plaintext is then $\mu = c_2 - \sum_i r_i \odot_d b_i$. For MP-CLWR, each inverse-rounding fiber is 1023 or 1024 consecutive residues, so a public midpoint lift L is at centered distance at most 512 from the private lift, and

$$\|v - v_0\|_{\infty, q} \leq 9 \cdot 257 \cdot 512 = 1184256 < q/8 = 2367702.875.$$

The reconciliation identity therefore returns the hash input, and one random-oracle query recovers the plaintext. The IND-CPA advantage is at least $(1 - \varepsilon^2)/2 = 0.499999865037 \dots$

The published stopping point is not where the attack stops. The bound $\varepsilon = (3^t - 1)/2q$ measures whether an endpoint table is *injective*, and at this modulus it exceeds 1 once $t \geq 16$, so beyond that it asserts nothing. The attack still works, because a collision does not defeat the recurrence: it forces a branch, and a wrong branch fails at the next layer, whose residual then has no preimage. The governing quantity is the expected number of spurious preimages per layer, $2^t/q$. Equivalently the recurrence must fix t bits per layer while each layer supplies $\log_2 q$ bits, so it collapses when 2^t approaches q .

Measured at the full published dimensions over three key generations per row, the true masks are recovered in 3/3 trials at $t = 9, 12, 16, 20, 22$ and 24, and in 0/3 at $t = 25$. Mean explored nodes rise from 258 at $t \leq 16$ to 2526 at $t = 24$, against a minimum of 258; the largest branch factor observed was 6. The collapse point matches $\log_2 q = 24.13$.

A second limit applies to every algorithm, and it turns out to be the schemes' own requirement in disguise. An IND-CPA adversary holds c_1, c_2 , and both candidate messages, so the observable ciphertext has $(n + k) + d = 1024 = k + d + n$ coefficients, exactly the quantity appearing in the two security lemmas. Counting bits, the masks stop being determined past $t^* = (k + d + n) \log_2 q / (k + 1) = 96.32$, while the lemmas

t	$2^t/q$	published $(3^t - 1)/2q$	recovered	mean nodes
9	0.0000	0.00052	3/3	258
16	0.0035	1.136	3/3	258
20	0.0554	92.0	3/3	273
22	0.2214	828	3/3	331
24	0.8857	7455	3/3	2526
25	1.7715	22370	0/3	capped

Table 5 Mask recovery against the number of mask polynomials, at the full published dimensions over three key generations per row. The published bound exceeds one from $t = 16$ and so asserts nothing beyond that point, while the attack runs to $t = 24$ and collapses at $t = 25$, where 2^t passes q .

require $t \geq (2\lambda + (k + d + n) \log_2 q)/(k + 1) = 97.3201$, hence 98. The two differ by exactly $2\lambda/(k + 1) = 0.9961$, one unit of t .

The randomness requirement is therefore not an incidental inequality: it says precisely that the mask entropy exceeds the ciphertext size by 2λ bits. That settles why the boundary attack cannot reach the theorem regime. It fails there because the ciphertext stops determining the masks, not because a table became large.

The scope is exact. Setting $\lambda = 128$ in the schemes’ own security lemmas requires $t \geq 98$, a factor 4.1 over the concrete attack and only 1.017 over t^* . No published parameter set uses $9 < t < 98$, so the extended range breaks no further printed row. What it changes is the margin, and what explains it. The reason usually given, the size of the 2^t table, is not the binding constraint; under-determination is. Bai et al. [3, Section 6.2.2] already treat mask recovery as an attack target, and the forward recurrence is the MPSign coefficient-layer attack [4] specialised to a binary subdomain. What is added here is the exact sign-class injectivity bound, the independent reverse endpoint, the near-one success probability over key generation, and plaintext recovery through both encryption interfaces.

4.4 A lattice e-voting protocol loses receipt-freeness

The printed fake-receipt simulator does not reproduce the genuine cumulative randomness law at any permitted ternary parameter, so a coercer who receives a transcript can read the vote and reject fabricated receipts.

Receipt-freeness is the property that a voter cannot prove to anyone else how they voted, even if they want to. It is what stands between a secret ballot and a market in votes, and it is usually obtained by letting the voter fabricate a receipt for any candidate: if a convincing fake exists, a genuine receipt proves nothing. The construction here provides such a simulator, and the attack is that its output is distinguishable from the real thing.

In the cast-as-intended protocol of Farzaliyev, Pärn, Saarse, and Willemson [10], the receipt carries the cumulative randomness $r_v = r_{vd} + r_s$ and the re-randomized ciphertext c^* , and public special decryption returns the claimed vote as $c^* - Ar_v$ with

no secret key. Privacy therefore rests entirely on the voter's ability to fabricate an indistinguishable receipt for a different vote.

Each ternary coefficient satisfies $\Pr[X = \pm 1] = p$ and $\Pr[X = 0] = 1 - 2p$, so a genuine cumulative coefficient is the convolution $Y = X_1 + X_2$ with

$$\Pr[Y = \pm 2] = p^2, \quad \Pr[Y = \pm 1] = 2p(1 - 2p), \quad \Pr[Y = 0] = 1 - 4p + 6p^2.$$

The printed simulator keeps the positions where the genuine value is ± 2 and replaces every other coefficient by a uniform element of $\{-1, 0, 1\}$, giving $\Pr[Y^* = \pm 2] = p^2$ and $\Pr[Y^* = -1] = \Pr[Y^* = 0] = \Pr[Y^* = 1] = (1 - 2p^2)/3$. The one-coordinate total variation distance is exactly

$$\Delta(Y, Y^*) = \frac{2(10p^2 - 6p + 1)}{3} = \frac{20}{3} \left(p - \frac{3}{10}\right)^2 + \frac{1}{15} \geq \frac{1}{15},$$

with equality only at $p = 3/10$. The bound is uniform over the permitted range $0 < p < 1/2$, so no choice of the ternary parameter repairs the simulator. This is the sharp form of the result: the defect is structural, not a parameter accident.

The gap is concentrated in the zero mass, $\theta_R = 1 - 4p + 6p^2$ against $\theta_F = (1 - 2p^2)/3$, so counting zero coefficients in one d -coefficient component suffices. At the published $d = 4096$ and $p = 1/3$, accepting a receipt as genuine when the zero count is at least 1210 gives exact binomial tails

$$\begin{aligned} \Pr[D(\text{fake}) = \text{genuine}] &= 1.046943719579 \cdot 10^{-7}, \\ \Pr[D(\text{genuine}) = \text{fake}] &= 9.114537336335 \cdot 10^{-8}, \end{aligned}$$

hence $\text{Adv}(D) = 0.999999804160$. For the centered-binomial marginal $p = 5/16$ the optimal cutoff 1235 gives 0.999997734193. One polynomial component suffices, so the attack does not depend on resolving the source's inconsistent use of d against $3d$.

Under the componentwise reading of the simulator across all $3d = 12288$ coefficients, the cutoff 3630 lowers the total error to $2.010188913363 \cdot 10^{-19}$. The one-component figure is the one to quote, because it does not depend on that reading.

The zero count is also close to the best possible statistic, and the exact sense in which it is best determines how the result can be attacked. The optimal advantage of any test is the total variation distance between the two product laws. Computing that distance exactly over the full multinomial and comparing it with the induced law of a candidate statistic shows that the pair

$$(\text{zero count}, \#\{\text{coefficients equal to } \pm 2\})$$

is sufficient: its distance equals the optimal one at every tested d and for both $p = 1/3$ and $p = 5/16$. The zero count alone is strictly weaker, for instance 0.187120914287 against the optimal 0.195640801881 at $d = 8$ and $p = 1/3$. So the exact optimal test thresholds the zero count conditionally on the observed ± 2 count, and no further statistic helps. The ± 2 cells carry no information alone, since the simulator reproduces them with probability p^2 exactly as the genuine law does; they matter only as the conditioning variable.

The complete coercer computes $v_T = c^* - Ar$, checks that it is the demanded candidate, and then applies the zero-count test to r . A genuine receipt is accepted and reveals the actual vote; the printed fake receipt is rejected. No lattice reduction, MLWE oracle, secret key, server interaction, or malicious verifier behaviour is required. BGV security, MLWE hardness, and ballot secrecy against an outsider holding no transcript are not attacked.

A repair must change the protocol or the simulation strategy, since reparameterisation is excluded by the universal bound. Candidate directions are a fully zero-knowledge re-randomization proof, sampling the fake cumulative randomness from the exact convolution law subject to every publicly visible constraint, or preventing a transferable party from holding both the randomness and the matching ciphertext.

4.5 A recovered permutation yields the old decryption key

The published Schur-product recovery of the hidden permutation in an updatable encryption scheme extends, by linear algebra alone, to recovery of a valid old decryption key and decryption of the pre-update challenge.

The Hollow-LWE updatable public-key encryption scheme of Albrecht, Benčina, and Lai [1] has old key pair $A^\top r = u$ with $r \in \{-1, 1\}^n$. An honest update applies a signed permutation $O = DP$ and a basis change U , publishing $H = DPAU$ and updating the secret to $r^* = DPr$. In the scheme's IND-CR-CPA experiment the adversary receives the old-key challenge ciphertext together with the updated public key and the exposed updated secret key.

Battagliola, Mora, and Santini [5] recover P , because squaring removes the signs:

$$(DPv) \star (DPw) = D^2 P(v \star w) = P(v \star w), \quad D^2 = I.$$

That result is theirs, it is prior public work, and it is not claimed here. Their analysis stops at the code-equivalence layer, leaving D apparently still hiding the key.

It does not. Write

$$B = PA, \quad H = DBU, \quad D = \text{diag}(d_1, \dots, d_n), \quad d_i \in \{-1, 1\}.$$

Choose k row indices $I = (I_1, \dots, I_k)$ with B_I invertible and set $V = BB_I^{-1}$ and $W = HH_I^{-1}$. Since $H_I = D_I B_I U$ we have $W = DVD_I$, so at every nonzero V_{ij} ,

$$\lambda_{ij} := W_{ij} V_{ij}^{-1} = d_i d_{I_j} \in \{-1, 1\}.$$

Form a graph on the row indices $[n]$ with an edge between i and I_j whenever $V_{ij} \neq 0$, labelled λ_{ij} . Within each connected component C , multiplying edge labels determines public signs $\delta_i \in \{-1, 1\}$ with $d_i = \varepsilon_C \delta_i$ for one unknown $\varepsilon_C \in \{-1, 1\}$ per component. Consistency around cycles is guaranteed by the existence of the true D . A component containing no index I_j consists of rows whose corresponding row of V is zero, and its sign does not affect B .

The component signs are then fixed by public data, which is the step that makes the recovery unconditional. The exposed updated secret satisfies $r^* = DPr$, so for a

choice of component signs write $\tilde{r}(\varepsilon)_i = \delta_i \varepsilon_{C(i)} r_i^*$, the true choice giving $\tilde{r} = Pr$. The old public-key relation $A^\top r = u$ is $B^\top \tilde{r} = u$, and since $B = VB_I$ this is $V^\top \tilde{r} = y$ with $y := B_I^{-\top} u$. Put $g_j := \sum_{i=1}^n V_{ij} \delta_i r_i^*$. If C_j is the component containing I_j , then $V_{ij} \neq 0$ forces $C(i) = C_j$, so $(V^\top \tilde{r}(\varepsilon))_j = \varepsilon_{C_j} g_j$. For every component meeting some I_j with $g_j \neq 0$, set $\varepsilon_C = y_j g_j^{-1}$; the true update relation guarantees this lies in $\{-1, 1\}$ and does not depend on which nonzero coordinate of the component is used. If all such g_j vanish, the same relation forces the corresponding y_j to vanish and either sign satisfies the equation. Components meeting no I_j are free.

This produces $\tilde{r}' \in \{-1, 1\}^n$ with $B^\top \tilde{r}' = u$, hence $r' := P^\top \tilde{r}' \in \{-1, 1\}^n$ with $A^\top r' = u$: a valid old decryption key. If the component signs are uniquely determined then $r' = r$; otherwise the public data admit several compatible signed secrets and any of them serves. For an old-key ciphertext $c_0 = Ax + e$ and $c_1 = \langle u, x \rangle + e_0 + \lfloor q/p \rfloor m$,

$$c_1 - \langle r', c_0 \rangle = e_0 - \langle r', e \rangle + \left\lfloor \frac{q}{p} \right\rfloor m,$$

because $A^\top r' = u$. Since $r' \in \{-1, 1\}^n$ carries the norm bound of an honestly generated key, the scheme's own correctness argument returns the challenge message.

A compatible signed update follows. With $D' = \text{diag}(\delta_i \varepsilon_{C(i)})$ and $U' = B_I^{-1} D'_I H_I$, the edge relations give $W = D' V D'_I$ including the zero entries, hence $H = D' B U'$. Permutation recovery therefore lifts by linear algebra to a valid signed update relation and an old decryption key, both publicly checkable against $H = DPAU$.

After the permutation is known the work is two $k \times k$ inversions, products of $n \times k$ by $k \times k$ matrices, graph propagation, and the public-key consistency equations: $O(k^3 + nk^2)$ field operations with classical matrix arithmetic. There is no sign search and no lattice reduction. If permutation recovery succeeds with probability ε_P , the IND-CR-CPA advantage is about $\varepsilon_P/2$.

Reduced-dimension runs over $\mathbb{F}_{3329}$ recovered the update relation, the originally sampled old key, and the challenge plaintext in 12/12 trials at each of $(n, k) = (64, 12)$, $(128, 16)$, and $(256, 24)$. A permutation altered by a single transposition was rejected in 12/12 trials. These runs validate the post-permutation algebra; they assume P and do not rerun the published full-size recovery. The argument above requires only a valid old decryption key; recovery of the originally sampled one is what these runs happened to observe.

A second defect is independent of that attack. The scheme's reduction transforms ordinary LWE of dimension $k - h$ into Hollow LWE of width k with hull dimension h , but the appended parameter script invokes the lattice estimator with $n = k$. The printed security labels are therefore not certified by the source's own reduction and estimator combination. This does not by itself place the rows below those levels; it shows the certification is computed at the wrong source dimension. A corrected search must estimate at $k - h$, raise k , recompute n , q , correctness, and h , and iterate, since changing k changes all of them.

Everything the published script does apart from the estimator call is arithmetic, and reproducing it recovers the source's parameter tables exactly, including n , $\log_2 q$, both components of h , and both size columns on all four rows. Two consequences follow, and they run against each other.

Repairing the dimension defect alone makes the code-equivalence exposure worse. It suffices to raise k until $k' - h' \geq k$, since the script already certified dimension k and hardness is monotone in the dimension, so that direction needs no estimator call. But $n = \lceil (1 + c)k \log_2 q \rceil$ grows with k while h does not, so every repaired row sits further inside the weak regime, with $\sqrt{2n}$ rising from 120.9 to 127.5 on the first row and similarly on the others.

Repairing both requires a fixed point, because raising h above $\sqrt{2n}$ forces k up to preserve $k - h \geq k_0$, which raises n and $\sqrt{2n}$ again. Iterating gives:

Row	$k \rightarrow k^*$	$h \rightarrow h^*$	n^*	ciphertext	update token
128, $p = 2$	450 $\rightarrow$ 600	27 $\rightarrow$ 140	9750	1.34 $\times$	1.33 $\times$
128, $p = 16$	550 $\rightarrow$ 750	26 $\rightarrow$ 179	15938	1.54 $\times$	1.54 $\times$
192	900 $\rightarrow$ 1150	37 $\rightarrow$ 228	25875	1.28 $\times$	1.28 $\times$
256	1250 $\rightarrow$ 1550	48 $\rightarrow$ 272	36813	1.24 $\times$	1.24 $\times$

Table 6 Cost of repairing both Hollow-LWE defects at once. Estimating at $k - h$ and leaving the weak Schur regime pull against each other, so the rows are the fixed point of iterating both.

The hull dimension has to grow by a factor between 5.2 and 6.9, and both ciphertexts and update tokens grow by 1.24 to 1.54 times. Update tokens were already the dominant cost, so the repaired scheme carries tokens between roughly 1 and 4.3 megabytes. These rows are a lower bound on the repair, since only the estimator-free direction is used.

4.6 A rerandomisable ciphertext answers whether its noise wrapped

Shifting one ciphertext coefficient coherently in both CRT limbs cancels in the plaintext until the exact-noise centre lift wraps, so the authenticated payload’s accept bit is a threshold oracle on the noise, and the noise determines the key.

The construction of Vodenicarevic et al. [9] recovers the exact noise during decryption: it computes $v_t = c_{0,t} - c_{1,t}s$ and $v_{q_2} = c_{0,q_2} - c_{1,q_2}s$, centre-lifts the q_2 limb to obtain ν , and returns $M = (v_t - \nu)\Delta_t^{-1} \bmod t$. Embedding the plaintext so that it vanishes modulo q_2 is what leaves that limb carrying only the noise, which the source states as the feature enabling exact recovery without rounding. Its parameters set $\Delta_t = q_2 \bmod t$, which is the join the attack turns on, because it forces

$$q_2 \Delta_t^{-1} \equiv 1 \pmod{t}.$$

Add the same integer δ to one coefficient of c_0 in both limbs, leaving c_1 untouched. While $\nu_j + \delta$ stays below $q_2/2$ the centre lift does not wrap, the two copies of δ cancel, and the plaintext is unchanged, so the inner authenticated payload verifies. As soon as the boundary is crossed the lift subtracts q_2 , and the identity above moves that one

plaintext coefficient by exactly one, so verification fails. The accept bit is therefore the monotone predicate

$$O_j(\delta) = [\delta < \lfloor q_2/2 \rfloor + 1 - \nu_j],$$

and binary search returns ν_j exactly. The q_2 limb carries no message, so once the whole noise polynomial is known $c_{1,q_2}s = c_{0,q_2} - \nu$ returns the secret wherever c_{1,q_2} is a unit; the secret is small, so centre-lifting its residue gives the integer polynomial and the other limb with it.

The source’s own scalar example reproduces this. At $t = 7$, $q_2 = 11$, $\Delta_t = 4$, $s = 2$, $M = 5$ and $\nu = 1$, the first rejecting shift is $\delta = 5$, which returns $\nu = 1$ and then $s = 2$. On a negacyclic ring at $n = 64$ with a 31-bit q_2 the attack recovers every noise coefficient and the secret from the accept bit alone, at 30 queries per coefficient. At the source’s $q_2 < 2^{32}$ and $n = 4096$ that is at most $4096 \cdot 32 = 131072$ adaptive reactions from a single captured ciphertext.

The mitigation the source proposes does not reach this. It recommends returning one generic failure indication so that a decryption oracle is not exposed, but the attack never asks why a decryption failed: accept against generic reject is the entire signal.

Not claimed. The IND-CPA and IND\$ statements for the raw layer. Those concern passive indistinguishability and this adversary is adaptive and active. What the attack contradicts is the composed claim that wrapping the malleable layer around an authenticated payload leaves a malicious rerandomiser with denial of service only. The attack is conditional on the acceptance being observable: a recipient exposing no difference between acceptance and failure admits no oracle, and a proof that every rerandomisation is a bounded, correctly formed encryption of zero would exclude the offsets. Both conditions are stated wherever the result is.

4.7 A perturbation rank is averaged where the attacker chooses

A MinRank target raised by perturbation is computed over an ambient dimension the attacker can publicly reduce, and at an average rank the attacker need not accept.

The scheme of Patarin, Vacek, and Roullet [8] hardens HFE by adding r perturbations $Q_i = L_i L'_i$, which raises the rank a MinRank attack must reach from d to about $d + r$. Three separate steps in that accounting do not hold. All figures below use the source’s own support-minor expression $C_{\text{MR}}(N, \mu) \approx \mu(N - 1)^4 \binom{2\mu+1}{\mu}^2$, so the comparison stays inside its cost model.

The Dragon variant writes its public matrices as $\tilde{H} = \begin{pmatrix} \alpha & \beta \\ 0 & 0 \end{pmatrix}$ and raises the ambient dimension from n to $n + m$ because the mixed block β adds columns. The split between signature and hash variables is public, so taking the principal aa block of every public matrix deletes β while keeping the transformed α . The target is unchanged and the dimension is n : the 128-bit row moves from $2^{131.89}$ to $2^{125.53}$, and the 80 and 100-bit rows likewise fall below their labels.

For the degree-3 variant the source polarises a cubic perturbation, finds a differential rank contribution of 0, 1 or 2 with probabilities $\frac{1}{8}, \frac{6}{8}, \frac{1}{8}$, and averages it to one per perturbation. An attacker picks the direction, so what matters is the lower tail of $S = X_1 + \dots + X_r$. Searching for a direction with $S \leq s$ and running MinRank at

$\mu = d + s$ costs $C_{\text{MR}}(n + m, d + s) / \Pr[S \leq s]$, and optimising s puts the 128-bit rows at $2^{100.75}$, $2^{102.22}$ and $2^{103.65}$ for $d = 2, 3, 4$, and the 256-bit row at $2^{187.45}$.

The signing distribution used for the concrete estimates sets exactly one Q_i to one. Every accepted signature then satisfies the degree-2 relation

$$1 + \sum_{i=1}^r L_i(x) L'_i(x) = 0,$$

where the source's own discussion of transcript structure reaches for degree-4 relations such as $Q_i Q_j = 0$. The polar form $B_R = \sum_i (u_i v_i^\top + v_i u_i^\top)$ has kernel $\bigcap_i \ker L_i \cap \ker L'_i$ when the forms are independent, so interpolating that one relation returns the common annihilator of the entire perturbation. Restricted to that kernel every LL' term vanishes identically and the target returns from $d + r$ to d : at the 128-bit rows the restored low-rank stage costs about $2^{38.5}$ and $2^{35.0}$. Interpolation needs enough transcripts to span the degree- ≤ 2 monomials, 32897 at $N = 256$ and 115441 at $N = 480$.

Not claimed. Any executed key recovery. These are statements inside the source's own cost model and about its own signing distribution; no MinRank attack was run at these parameters, and the restored low-rank figures are the cost of that stage rather than of a completed recovery. The open bridge is to feed the restricted instance through an HFE MinRank implementation. A signer that mixes odd and even hypothesis weights destroys the constant relation, so the exact-one tables and the randomised-weight proposal are not the same scheme.

4.8 An explicit normal basis splits along the CRT factors

The constructive normal-basis generator of Proposition 6.11 has a Galois orbit supported factorwise, so the stated degree-63 instance decomposes into seven degree-nine instances at an unchanged noise rate.

Bombar, Couvreur, and Debris-Alazard [7] build a normal-basis generator inside the CRT product as $b = (a, 0, \dots, 0)$ and take its complete Galois orbit. Because the generator is supported on one component, so is every element of the orbit. The global basis is therefore a disjoint union of local normal bases instead of a genuinely global one.

The consequence for the noise is exact. Error sampled as $e = \sum_{\sigma \in G} e_\sigma \sigma(b)$ with independent Bernoulli coefficients projects onto a component by annihilating every basis element supported elsewhere. The surviving error is a local Bernoulli error with the *same* parameter: there is no compensating increase. Measured over 100000 trials at source rate $1/4$, the seven per-factor rates all lay within 0.0006 of the source rate.

For the stated example, exhaustive trial division over $\mathbb{F}_2$ gives

$$X^{63} + X^7 + 1 = \prod_{i=1}^7 f_i(X), \quad \deg f_i = 9,$$

with seven distinct irreducible factors and residual cofactor 1. Hence, for this basis,

$$\text{NormalRingLPN}_{63} \longrightarrow 7 \times \text{RingLPN}_9.$$

Each local secret has only $2^9 = 512$ candidates, so the complete CRT secret follows from seven independent maximum-likelihood searches using $7 \cdot 512 = 3584$ candidate tests per scan, against 2^{63} undecomposed.

Scope requires care here. The authors themselves warn that their constructive basis can cancel noise under projection and suggest that the normal basis need not be the one they construct. That warning is prior public information and is not claimed here. What is added is the exact basis-support decomposition, the observation that the projected rate is unchanged and not merely reduced, the precise $63 \rightarrow 7 \times 9$ collapse, and the resulting candidate count. An independently sampled random normal basis is untouched, and the search-to-decision theorem is not contradicted: it can remain true while both search and decision are easy under this particular basis.

4.9 Semidirect group-ring LWE has two public maps

The augmentation defeats the full-ring decision claim, and explicit rank-four maps defeat the selected quotient rings under their coefficient Gaussian completion; the two maps are complementary, since the augmentation does not descend to either quotient.

Liu and Fu [13] study LWE over full and quotient group rings for two semidirect families, selecting quotients intended to remove the one-dimensional representations that enable the simplest homomorphism attacks.

Their decision statement is ill-typed before any attack: the definition asks for a probability distribution over error distributions, while the theorem supplies only the set $\Psi_{\leq \alpha}$ of elliptical Gaussians. The analysis below fixes an explicit coefficient-basis Gaussian completion and attributes no formal quotient statement to the source.

The full rings.

Every integral group ring carries the augmentation

$$\varepsilon\left(\sum_{g \in G} c_g g\right) = \sum_{g \in G} c_g,$$

a surjective ring homomorphism onto $\mathbb{Z}$. A genuine sample $b = sa/q + e \bmod R$ projects to the one-dimensional sample $B = SA + E \bmod q$ with $S = \varepsilon(s)$, $A = \varepsilon(a)$, and $E = q\varepsilon(e)$, while a uniform sample stays uniform. Since ε sums $N = |G|$ independent coefficients, E has width at most $\alpha q \sqrt{N}$.

Take the sequence permitted by the source's own numerical condition:

$$N^{10} < q_N < 2N^{10}, \quad \alpha_N = \frac{2N}{q_N}, \quad B_N = 4N^{3/2} \log N,$$

so that $\alpha_N q_N = 2N$. Given two samples, enumerate $S' \in \mathbb{Z}_{q_N}$ and accept when both residuals are at most B_N in absolute value. The genuine secret passes by Gaussian concentration, since the error width is at most $2N^{3/2}$ and the ratio of $B_N/2$ to it is

$\log N$. For the uniform distribution, each candidate passes with probability at most $(2B_N/q_N)^2$, so a union bound over q_N candidates gives

$$q_N \left(\frac{2B_N}{q_N} \right)^2 = \frac{4B_N^2}{q_N} = 64 N^{-7} \log^2 N.$$

Enumeration costs $q_N = N^{10+o(1)}$, which is polynomial in the encoded instance size. The natural universal repair of the full-ring decision claim is therefore false: there is an explicitly permitted coefficient-Gaussian sequence for which the problem is polynomial-time distinguishable.

At the stated parameter sequence, the two-sample test accepted the genuine distribution 40/40 and the uniform distribution 0/40 for Type I with $m = 4$ and $m = 6$ and Type II with $k = 1$ and $k = 2$. The one-sample control accepted uniform 40/40, confirming that the second sample is load-bearing.

The quotients.

The augmentation does not reach them. Exact computation gives $\varepsilon(N_k) = 3^k$ for the Type-II norm quotient and $\varepsilon(t^2 + 1) = 2$ for Type I, so ε survives only modulo 3^k and modulo 2 respectively. The selected quotients require separate maps, and they exist: for Type I with $n = 4$, reduction of the exponent of s modulo two gives a rank-four quotient; for Type II, coordinate reduction modulo three descends to a ring epimorphism $R_k \twoheadrightarrow R_1$ because $N_k \mapsto 3^{k-1}N_1$. Both targets satisfy $R_1/qR_1 \cong M_2(\mathbb{F}_q)$ for the attack primes. Enumerating the rank-four quotient secret and testing two projected residuals distinguishes in $N^{40+o(1)}$ operations with uniform false acceptance $N^{-28+o(1)}$.

For Type I the scope of this quotient mechanism can be made exact. Since $t^2 = -1$ gives $t^{-1} = -t$, the defining relation $sts^{-1} = t^{-1}$ becomes $st = -ts$; write $R_{m,q}^I$ for the resulting rank- $2m$ algebra modulo q , and take m even, q odd, and $q \equiv 1 \pmod{m}$, which Dirichlet permits inside the parameter sequence. For each m -th root of unity $z \in \mathbb{F}_q^\times$,

$$\rho_z(s) = \begin{pmatrix} z & 0 \\ 0 & -z \end{pmatrix}, \quad \rho_z(t) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

satisfy $\rho_z(s)^m = I$, $\rho_z(t)^2 = -I$, and $\rho_z(s)\rho_z(t) = -\rho_z(t)\rho_z(s)$, so ρ_z is a homomorphism $R_{m,q}^I \rightarrow M_2(\mathbb{F}_q)$. The representations for z and $-z$ are conjugate, and taking one representative from each pair $\{z, -z\}$ gives

$$R_{m,q}^I \cong \prod_{z \in \mu_m(\mathbb{F}_q)/\{\pm 1\}} M_2(\mathbb{F}_q).$$

Injectivity is direct: every element is uniquely $x = A(s) + B(s)t$ with $\deg A, \deg B < m$, and

$$\rho_z(x) = \begin{pmatrix} A(z) & -B(z) \\ B(-z) & A(-z) \end{pmatrix},$$

so if all blocks vanish then A and B vanish at all m roots of $X^m - 1$ and are zero. Source and target both have dimension $2m$ over $\mathbb{F}_q$, so the map is an isomorphism. Verified exactly for $m = 2, 4, 6, 8, 10, 12, 16, 20$.

That decomposition turns the recorded limitation into a bound. For $x = \sum_{i < m} (a_i s^i + b_i s^i t)$ write $\|x\|_{\text{coef}, \infty} = \max_i \{|a_i|, |b_i|\}$, and let $\|\cdot\|_{\infty, q}$ be the largest absolute centred representative of a matrix entry. For the block $z = 1$ every entry is a signed sum of m coefficients, so

$$\|\rho_1(x)\|_{\infty, q} \leq m \|x\|_{\text{coef}, \infty}.$$

Every other block distorts far more. Let $z \notin \{\pm 1\}$ have multiplicative order d . Then

$$\max_{0 \leq j < d} |[z^j]_q| \geq \sqrt{\frac{q}{2}},$$

where $[\cdot]_q$ is the centred representative. Otherwise every element of $\langle z \rangle$ would have centred representative below $\sqrt{q/2}$ in absolute value, the product of any two would stay below $q/2$, and multiplication in the subgroup would agree with integer multiplication without wraparound. The centred representatives would then form a finite multiplicative subgroup of the nonzero integers, which is contained in $\{\pm 1\}$, contradicting $z \notin \{\pm 1\}$. Choosing j attaining the bound, the basis element s^j has $\|s^j\|_{\text{coef}, \infty} = 1$ while $\rho_z(s^j) = \text{diag}(z^j, (-z)^j)$ has $\|\rho_z(s^j)\|_{\infty, q} \geq \sqrt{q/2}$.

Because the split algebra is a direct product of simple matrix rings, every two-sided quotient retains a subset of the blocks. So when $m < \sqrt{q/2}$, the rank-four $z = 1$ image is maximal among quotient images whose coefficient-to-centred distortion stays below $\sqrt{q/2}$: any strictly larger quotient contains a nontrivial block and loses that property. This is the maximality the bounded-residual attack needs, and it is not the claim that every statistical attack on the other blocks is impossible. Measured at $m = 20$ and $q = 100361$ over 400 errors with coefficients in $\{-1, 0, 1\}$, the largest centred entry was 12 for the $z = 1$ block against $|G| = 80$, and between 48253 and 50167 for each of the nine others against $q/2 = 50180$, consistent with the two bounds. A repair must remove that single trivial-character direction.

Both results recover only the image of the secret under the respective map, and neither instantiates an attack on a fully specified encryption scheme. The source supplies no Type-II parameter generator or message encoding. Standard RLWE is not affected. Removing one-dimensional representations is, on this evidence, insufficient: what must be excluded is every efficiently computable image in which the secret can be enumerated and the projected error remains distinguishable.

4.10 An unrestricted hint definition is not generically hard

Read without the restrictions its hardness theorem imposes, the Hint-MLWE definition admits a parameter choice that reveals the randomness exactly.

The parameterized definition of Kim, Lee, Seo, and Song [12] admits $\ell = 1$, $\gamma = 1$, and a point-mass hint distribution. That choice makes the hint reveal the randomness exactly and gives a distinguisher of advantage $1 - q^{-nm}$.

This does *not* contradict the source’s Gaussian hardness theorem, which excludes that parameter choice. It bounds how the assumption may be cited: a construction may invoke only the restricted family the theorem covers, not “Hint-MLWE” generically.

5 Where a proof covers less than the construction needs

The three targets in this section behave differently from everything above. Nothing in them is wrong, and nothing in them is attacked.

In each case the pattern is the same. A paper proves that some problem is hard under a stated set of conditions. A construction is then built on the same assumption, but relies on it holding in a wider range than the proof covers. Nobody closes that gap, and usually nobody claims to: the authors are often explicit about what their theorem does and does not reach. The difficulty arrives later, when the assumption is cited by name, because the name carries the wide reading while the proof only supports the narrow one.

So what these three entries support is a statement about how an assumption may be cited, not a statement about whether a scheme is secure. Presenting them alongside the findings above would make this record claim more than its evidence does, and calling them breaks would be simply false. They are here so that the next reader does not have to rediscover the difference, and separate so that nobody mistakes the difference for an attack.

Sparse LWE.

Jain, Lin, and Saha [11] derive hardness from ordinary LWE in dimension about the sparsity k , so the reduction reads $\text{LWE}_k \lesssim \text{SparseLWE}_{n,k}$ and not $\text{LWE}_n \lesssim \text{SparseLWE}_{n,k}$. Raising n at fixed k does not inherit the hardness normally associated with dimension n , and a blockwise search at constant k is polynomial. The authors explicitly study constant and polylogarithmic sparsity and present this picture themselves, so this is not recorded as a finding against them. The defensible statement is about citation, not correctness: security is governed by k .

Tensor LWE.

Agrawal, Rossi, Yadav, and Yamada [2] reduce standard LWE to Tensor LWE for $x_i = 0$ and for one common fixed x , and explain why the argument does not extend to several distinct x_i sharing one secret. The ABE construction nevertheless invokes the full varying- x_i assumption, and its theorem is additionally very selective. A cancellation direction using public λ_i with $\sum_i \lambda_i r_i = 0$ and $\sum_i \lambda_i r_i[j] x_i[h] = 0$ would cancel the secret-dependent terms, but requires a relation short enough that $|\sum_i \lambda_i e_i| \ll q$, which is itself an SIS-type problem. No complexity bridge was found.

Succinct LWE.

Wee [17] gives an ordinary-LWE argument for the wide regime $\hat{m} = \ell m$, while the construction that provides the headline compression uses $\hat{m} = m$, supported by adding public-coin evasive LWE. The succinctness gain lives precisely in the part of the assumption that ordinary LWE does not establish. Turning the public trapdoor into a short dual relation requires a short vector in $\ker T_W \cap \ker(I_\ell \otimes G)$, again an SIS problem, and the source itself discusses the same danger in the extreme narrow regime.

None of the three is a reason to distrust the paper it comes from. Each is a reason to check, before relying on one of these assumptions, which version of it the cited theorem actually establishes.

6 Reproducibility

Every check below uses only the Python standard library, runs in seconds, and contains at least one discriminating control. A control fixes its expected outcome before running and shows that the apparatus separates the hypothesised phenomenon from a neighbouring case; whether it is required to pass or to fail is secondary. Table 7 lists what each check establishes and what its control is.

Two limitations apply. The Hollow-LWE runs assume the permutation and validate only the post-permutation algebra at reduced dimensions; they are not a benchmark of the full-size recovery, which is other authors' work. The corrected Hollow-LWE parameter rows are a lower bound on the repair, because only the estimator-free direction is used.

For Hollow-LWE, Tables 1 and 2 of [1] supply the row values and hull split, and the appended script fixes the estimator call. The permutation-recovery premise comes from [5]; Section 4.5 supplies the bridge to old-key and plaintext recovery.

7 What a security argument should enumerate

Together, the case studies show that autonomous search can discover reproducible failures at representation and distribution boundaries before human review.

Each defect arises before the advertised hard problem becomes relevant. They required reading what each construction publishes, asking which efficiently computable maps survive, and checking whether each asserted distribution is the one that actually occurs. Every result here survived because a short deterministic program with a control confirmed it, and several plausible candidates did not survive that step.

The recommendation is narrow. When a construction introduces a new algebraic setting or representation layer, the security argument should enumerate the efficiently computable images of its published objects and state, for each, why the secret is not enumerable there and why the projected error is not distinguishable. An input encoding must additionally be shown injective or uniquely decodable before security properties are transferred through it. Removing one class of maps, as the semidirect quotients remove one-dimensional representations, is not sufficient. Separately, when a verifier accepts a set of values, binding and soundness must be proved for that exact set and not for the honest element of it, and when a parameter script certifies a security level, it must be invoked at the dimension the reduction actually supplies.

AI disclaimer. In this paper about AI use, AI tools were used

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Check	Establishes	Control, with its required outcome
Ring-LWR opening	Universal double opening, the witness-free prover, the false-statement density, and the nonunit challenge difference	Honest opening with multiplier one still verifies
Spinel encoding	The two collisions, the one-byte second preimage, the 244/255 count, the exhaustive two-byte enumeration, and the SL_4 digest space	The fixed-width six-trit repair must separate every colliding pair, and the per-byte codeword map must be injective
Ring-LWR extraction density	The unit criterion, and that extraction is undefined for $\frac{1}{2} - 1/(2^{257} - 2)$ of challenge pairs	Brute-force inversion on two small rings must agree with the parity criterion
CRT-RLWE reaction	The source's scalar example, and full noise plus key recovery on a ring at $n = 64$ from the accept bit alone	Honest decryption must reproduce the plaintext, and a single-limb shift must never preserve it
MinRank accounting	The projected dimension, the optimised lower tail, and the restricted low-rank stage, all under the source's own expression	The per-perturbation rank law must reproduce the source's average of one
Middle-product boundary attack	All 2313 masks and both plaintexts at the full published dimensions, and the midpoint-lift bound	The reconciliation bound is checked, not assumed
Middle-product range	Recovery at $t \leq 24$ and collapse at $t = 25$, three key generations per row	$t = 25$ must exhaust the node budget
Receipt simulator	The exact distance, the $1/15$ bound, the binomial tails, and sufficiency of (zero count, $\# \pm 2$)	A simulator drawn from the correct convolution law must show no advantage
Hollow-LWE bridge	A valid old key recovered, and the challenge decrypted, 12/12 at three dimensions	A permutation altered by one transposition must be rejected
Hollow-LWE parameter repair	Reproduction of the source's own two tables, and the cost of repairing both defects	The reproduction must match every published row before anything is built on it
Constructive basis collapse	The exact seven-way degree-nine splitting and the unchanged projected noise rate	A perturbed factor must break the product, and a neighbouring trinomial must not split the same way
Group-ring augmentation	Multiplicativity, the union bound, separation at the stated parameters, and the count of sign characters	One sample instead of two must fail to separate
Type-I block maximality	The complete splitting, the m distortion bound for the trivial-character block, and the $\sqrt{q/2}$ lower bound for every nontrivial block	The $z = 1$ block must stay below the m bound; every nontrivial block must contain a basis witness of centred norm at least $\sqrt{q/2}$

Table 7 Each executable check, what it establishes, and its control.

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A The grind calculus

Section 1 states the method in prose. The calculus below states it in notation, fixing which discipline each step enforces and where a proposal becomes evidence. Admission assigns the intermediate judgement ev ; the classification rules below assign a final result type after human audit. The two axioms carry the weight.

A.1 The arena

Definition A.1 (Arena) Let Π be a set of source documents. Each $\pi \in \Pi$ carries a *version stamp* $\nu(\pi)$, meaning a revision date together with an archive identifier. The predicate $\text{Ver}(\pi)$ holds when the source version under attack is recorded, which is needed only because a preprint may be revised after the attack and a reader needs to know which text was read. Write

$$\mathcal{J}(\pi) = \{\text{joins of } \pi\},$$

the finite set of points at which the argument of π crosses a boundary: a change of algebra, a representation or encoding change, a distribution carried through a map, a rounding or CRT conversion, a quotient, an exceptional value, a quantifier widened from one object to many, or a parameter substituted into a script instead of into its theorem. The *hypothesis space* $\mathcal{H}$ consists of triples

$$\eta = (\gamma_\eta, \rho_\eta, \omega_\eta),$$

a claim γ_η of π , a mathematical or state relation ρ_η alleged to violate it, and a predicted observable ω_η fixed *before* anything is run.

Definition A.2 (Search directions) $\mathfrak{D}$ is not turned loose on $\mathcal{J}(\pi)$ unaimed. At each join j it is put to two questions, written A and B:

A(j) : which efficiently computable ϕ survives j with useful signal,

B(j) : which law actually results from j .

These generate $\mathcal{H}_A(j)$ and $\mathcal{H}_B(j)$, and $\mathcal{H} = \bigcup_j (\mathcal{H}_A(j) \cup \mathcal{H}_B(j))$. Section 3 states the same two questions in prose. A hypothesis in $\mathcal{H}_A(j) \cap \mathcal{H}_B(j)$ answers both at one join, and those are the ones that reach the highest levels of Λ .

A.2 The agents

Five agents act on the arena. They are written in fraktur because they are not functions in any respectable sense; $\mathfrak{D}$ in particular is a sampler whose distribution nobody can write down, and which is steerable only through the join it is given and the two questions asked there.

Agent	Type	Duty
$\mathfrak{D}$ (<i>the Dreamer</i>)	$\mathcal{J}(\pi) \rightarrow 2^{\mathcal{H}}$	Proposes many η per join. Optimistic, fluent, usually wrong
$\mathfrak{X}$ (<i>the Executor</i>)	$\mathcal{H} \rightarrow \{\top, \perp, ?\}$	Runs the exact check and its control. Returns ? for timeout, cancellation, or tool failure
$\mathfrak{E}$ (<i>the Executioner</i>)	$\mathcal{H} \rightarrow \{\perp\}$	Retires η with the discriminating evidence attached. Never deletes
$\mathfrak{P}$ (<i>the Pusher</i>)	$\mathcal{H} \rightarrow \mathcal{H}$	Sends a survivor with limitation ℓ to $\partial_\ell \eta$, the hypothesis that ℓ is removable
$\mathfrak{S}$ (<i>the Scribe</i>)	$\mathcal{H} \rightarrow \Xi$	Commits verdicts to the evidence store Ξ , including the rejected ones

Table 8 The five agents of the autonomous loop, their types, and what each is responsible for. Only $\mathfrak{X}$ decides anything; the rest propose, retire, re-aim, or record.

The five agents execute the autonomous loop. Final classification is performed by a human auditor through the predicate *Aud* defined below.

Axiom A.3 (Confident Nonsense) *Let $\kappa(\eta)$ be the confidence $\mathfrak{D}$ attaches to η , however expressed. Then κ is inadmissible: no rule below may mention it, and no ranking, ordering, or triage used to reach a judgement may be derived from it.*

Axiom A.3 is the whole reason the calculus exists, and it is stated as an exclusion on purpose. Calibrating κ against the truth would require an empirical claim about a particular generator at a particular time, which would date immediately and would have to be redone for the next one. Excluding κ requires nothing of the generator at all. A proposal that is wrong at random is harmless; a proposal that is wrong *fluently* will otherwise supply its own peer review.

Generator confidence may order the queue but does not enter admission or classification. Under a finite budget, record the scheduling rule and every generated candidate left unevaluated, since scheduling determines coverage.

A.3 Predicates

The rules quote the conditions in Table 9. Each is a property of a single hypothesis $\eta \in \mathcal{H}$, except *Brg*, which relates a hypothesis to a security property P . They divide into three jobs. *Ex*, *Wit*, *Ctl* and *Par* decide whether η becomes evidence at all. *Prm*

and Brg decide whether evidence reaches a security property. Clm, Nar and Att decide which type it receives, and Aud is the human step that every classification rule requires.

Two symbols need separating, because they look alike and do opposite jobs. Throughout, η is the hypothesis under test, and $\bar{\eta}$ is its *control*: a second hypothesis, built to differ from η in the one respect the analysis claims is doing the work, and run through the same apparatus. The pair exists so that the apparatus can be shown to tell them apart.

Predicate	Holds when
$\text{Ex}(\eta)$	Every decision on the path to ω_η is taken in exact rational or integer arithmetic. Floating point may inform, never decide
$\text{Wit}(\eta)$	A minimal deterministic witness reproduces ω_η
$\text{Ctl}(\eta)$	A control $\bar{\eta}$ exists with its outcome $\mathfrak{X}(\bar{\eta})$ fixed before it runs, separating η from a neighbouring case; the check is void if $\bar{\eta}$ returns anything else
$\text{Par}(\eta)$	The witness was run at the parameter sequence the analysis specifies, not at a convenient one
$\text{Prm}(\eta)$	The trigger uses only values the target itself permits
$\text{Brg}(\eta, P)$	A validated bridge carries ρ_η to an observable violation of P
$\text{Clm}(\eta)$	ρ_η falsifies a claim the source states, or leaves it unsupported
$\text{Nar}(\eta)$	What the source proves is materially narrower than what a consumer of it relies on, with no stated claim falsified
$\text{Att}(\eta)$	The relation is due to other authors
$\text{Aud}(\eta)$	The handoff contains the exact claim and source version, witness, control, code, environment, run record, limitations, and proposed type, and a human auditor has verified the source, consequence, scope, and attribution

Table 9 The predicates the rules quote. η is the hypothesis under test, $\bar{\eta}$ its control, ω_η its predicted observable, ρ_η the relation it alleges, and P a security property.

Ctl is a demand for *discrimination*, not for a particular verdict. A negative control that must fail shows the apparatus can register a negative; a positive control that must pass shows it reproduces a case already known to hold. Both appear in Table 7. What is excluded is a check whose outcome would have been accepted either way.

A.4 Evidence types, and the axiom that keeps them apart

Definition A.4 (Evidence types) The type of a survivor is drawn from

$$\mathbb{T} = \{ \text{brk}(P), \text{def}, \text{gap}, \text{kno} \},$$

a break of property P , an assumption or certification defect, a coverage gap, and a known correction attributable to other authors.

Axiom A.5 (No Coercion) *Regard $\mathbb{T}$ as a category whose objects are the four types. Its only morphisms are the identities. In particular there is no arrow $\text{def} \rightarrow \text{brk}$, no arrow $\text{gap} \rightarrow \text{def}$,*

and no composite producing one. A promotion may be obtained only by deriving the target judgement from scratch through Rule A.8.

Rule A.6 (Counting) *Let $\#$ be the tally measure on Ξ . Then $\# \text{kno} = 0$: results attributable to others are cited at the point of use and contribute nothing to the count, however load-bearing they are. The Schur-product permutation recovery that Section 4.5 builds on is the case that matters here.*

A.5 Admission and promotion

Rule A.7 (Admission)

$$\frac{\text{Ver}(\pi) \quad \text{Ex}(\eta) \quad \text{Wit}(\eta) \quad \text{Ctl}(\eta) \quad \text{Par}(\eta)}{\Xi \vdash \eta : \text{ev}} \text{ (ADM)}$$

The conclusion $\Xi \vdash \eta : \text{ev}$ records a reproduced candidate; Rule A.8 assigns its cryptanalytic type.

Rule A.8 (Classification)

$$\begin{array}{c} \frac{\text{Aud}(\eta) \quad \text{Att}(\eta)}{\Xi \vdash \eta : \text{kno}} \text{ (ATT)} \quad \frac{\Xi \vdash \eta : \text{ev} \quad \neg \text{Att}(\eta) \quad \text{Aud}(\eta) \quad \text{Prm}(\eta) \quad \text{Brg}(\eta, P)}{\Xi \vdash \eta : \text{brk}(P)} \text{ (BRK)} \\ \frac{\Xi \vdash \eta : \text{ev} \quad \text{Clm}(\eta) \quad \text{Aud}(\eta) \quad \neg \text{Brg}(\eta, P) \text{ for every } P}{\Xi \vdash \eta : \text{def}} \text{ (DEF)} \\ \frac{\Xi \vdash \eta : \text{ev} \quad \text{Nar}(\eta) \quad \text{Aud}(\eta) \quad \neg \text{Clm}(\eta)}{\Xi \vdash \eta : \text{gap}} \text{ (GAP)} \end{array}$$

Absence of a bridge does not on its own make a defect, which is why DEF carries Clm. A validated relation that falsifies nothing the source states and narrows nothing a consumer relies on receives no type at all: it is an observation, it stays in Ξ , and it is not a finding. ATT takes precedence over the other three, so a result belonging to other authors cannot be relabelled by finding a bridge for it.

The proposed type in the handoff is not a premise of Rule A.8.

The four rules are deliberately not exhaustive. Every type on the left of the turnstile has a premise that must be established by argument, and a hypothesis meeting none of them is left unclassified rather than pushed into the nearest available box.

Rule A.9 (Unresolved)

$$\frac{\mathfrak{X}(\eta) = ?}{\Xi \vdash \eta : \text{unresolved}} \text{ (UNR)}$$

A timeout, an out-of-memory kill, a cancellation, or a tool failure is an unresolved evidence path. It is never a refutation of η , and the temptation to read it as one is the reason the rule is written down.

Rule A.10 (Push)

$$\frac{\Xi \vdash \eta : \text{brk}(P) \text{ with stated limitation } \ell}{\partial_\ell \eta \in \mathcal{H}} \text{ (PUSH)}$$

Every limitation re-enters the hypothesis space as a target.

An admitted candidate may be refined before audit:

$$\frac{\Xi \vdash \eta : \mathbf{ev} \text{ with stated limitation } \ell}{\partial_\ell \eta \in \mathcal{H}} (\text{AutoPush}).$$

Thus autonomous search can attack a candidate's stated limitations without assigning it a final type.

Lemma A.11 (Two Exits) *A completed push has two productive exits. Either $\partial_\ell \eta$ is admitted, and the result strengthens; or $\partial_\ell \eta$ is refuted by a proof that ℓ cannot be removed, and ℓ upgrades from an apology to a maximality theorem. A push need not complete: by Rule A.9 it may return ? and stand unresolved, with neither a stronger attack nor an impossibility proof in hand. That is the ordinary outcome and is recorded as such.*

Lemma A.11 is why the push is worth running even when it is expected to fail. The Type-I block maximality statement in Section 4.9 is precisely a second exit: an attempt to widen a quotient attack that instead proved the quotient could not be widened.

A.6 The grind operator

Definition A.12 (Grind) Let $\Sigma_n \subseteq \mathcal{H}$ be the survivors after round n and Ξ_n the evidence store. The grind operator is the composite

$$\Gamma = \mathfrak{S} \circ \mathfrak{P} \circ (\mathfrak{X} \sqcup \mathfrak{E}) \circ \mathfrak{D}, \quad (\Sigma_{n+1}, \Xi_{n+1}) = \Gamma(\Sigma_n, \Xi_n).$$

A run reaches *grind normal form* at the least n with $\Sigma_{n+1} = \Sigma_n$: every surviving limitation has been attacked and has moved, been proved immovable, or been left unresolved with the attempt recorded. Normal form is a stopping condition, not a completeness claim.

Thus Γ governs autonomous search; Aud and Rule A.8 act on its output.

What survived this run is countable: eleven results, three coverage gaps, and one rejected strengthening retained for its discriminating evidence. What was proposed and failed was not counted, because the generator was never instrumented to record it.

That asymmetry is the economics of the method rather than a gap in its reporting. The survival rate could be made arbitrarily small by proposing more, and the method would be no worse for it, because the cost of a failed hypothesis is bounded by the cost of one exact check while the value of a surviving one is not bounded at all. What matters is the size of Σ_∞ , not the fraction of proposals reaching it.

A.7 Scope of the calculus

The calculus fixes two layers: where hypotheses come from, through the joins and the two search directions, and when a hypothesis is admitted as computational evidence and how an audited hypothesis is classified. It leaves a third layer alone. Which joins in a given construction repay attention, and in what order, is not derivable from anything above, and a run that picks badly returns nothing while satisfying every rule here. $\mathfrak{D}$ supplies volume and A and B supply aim, and neither supplies judgement about where to point them first.

A.8 Formal workflow and escalation ladder

The two diagrams of Section 1 are repeated here with every arrow named. Figure 3 is the loop of Figure 1 with the agents, the admission premises, and the three verdicts labelled, so that the door into evidence and the door into a classification appear as the separate steps they are. Figure 4 is the ladder of Figure 2 as a chain in the escalation poset, carrying the fibres that the earlier figure deliberately leaves out: there the rungs matter and here their contents do.

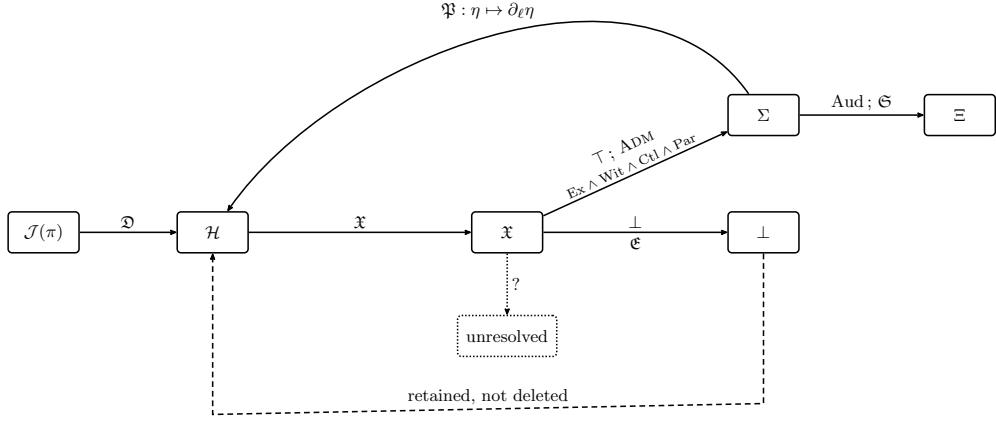


Fig. 3 Figure 1 with the arrows named. $\mathcal{D}$ proposes at the joins, ADM records a reproducible candidate and Aud precedes cryptanalytic classification, and $\mathfrak{P}$ returns every stated limitation to $\mathcal{H}$. The $\perp$ branch is archived and not discarded, so a rejected hypothesis keeps its discriminating evidence and can be reopened. The third verdict is a sink: an unresolved path is neither evidence nor a refutation, and nothing may be concluded from reaching it.

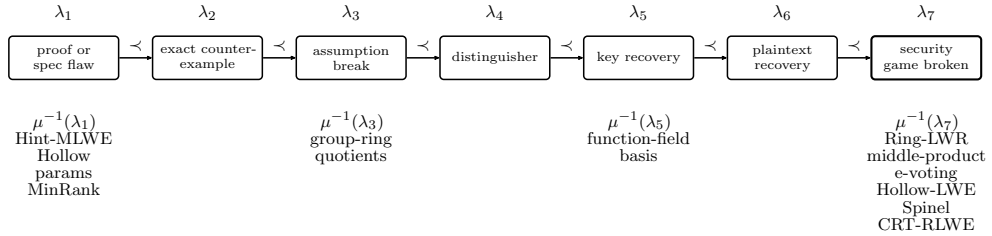


Fig. 4 Figure 2 as a chain $\lambda_1 < \dots < \lambda_7$ in the escalation poset Λ , with the attainment map $\mu : \Sigma_{\infty}^{\text{Aud}} \rightarrow \Lambda$, where $\Sigma_{\infty}^{\text{Aud}} = \{\eta \in \Sigma_{\infty} : \text{Aud}(\eta)\}$, sending each audited survivor to the highest level its evidence reaches. The fibres $\mu^{-1}(\lambda_i)$ are the contents of each level. A finding halts at $\mu(\eta) = \lambda_i$ exactly when $\neg \text{Brg}$ holds for every P at λ_{i+1} , or when Lemma A.11 has taken its second exit and the halt is itself a theorem.